

KnockKnock

Low-Cost IoT Intrusion Detection for Doors and Windows

IoT: Algorithms and Services

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Objectives

1 Objectives

- Low-power intrusion detection for doors and windows
- Reduce energy via hardware (ESP32-C3, TLC393) and duty-cycle design
- Compare connectivity: ESP-NOW vs BLE
- Show sensor-to-hub pipeline with vibration detection and alerting



Architecture Overview

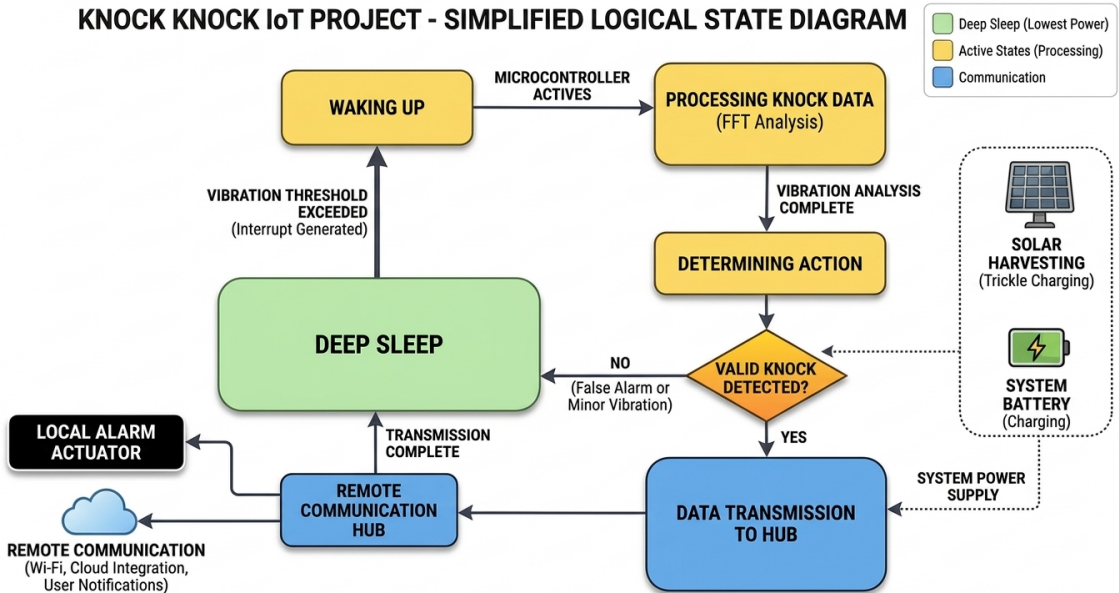
2 Architecture

- **Two-tier** network: edge nodes + central hub
- Edge nodes: piezo sensor + LM393 + ESP32 V1
- Hub: always-on ESP32, **[TODO]** bridges to cloud via Wi-Fi
- Communication: **ESP-NOW** (node → hub)
- Cloud: **[TODO]** **MQTT** broker + web dashboard

Design Principle

Sense and classify at the edge, act at the hub.

KNOCK KNOCK IoT PROJECT - SIMPLIFIED LOGICAL STATE DIAGRAM





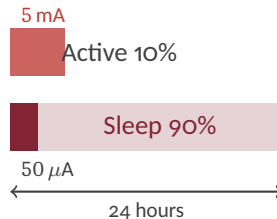
Duty Cycle

3 Energy & Duty Cycle

- **ESP32-C3 active:** 20 mA
- **Peak current (WiFi / ESP-NOW TX):** 200 mA
- **Alarm rate:** 12 alarms/hour = 288 alarms/day
- **Active time per alarm:** 30 s $\Rightarrow$ 86 400 s/day
total wake time (worst case: 1 alarm/s)
- **Duty cycle:** 10%

Key principle

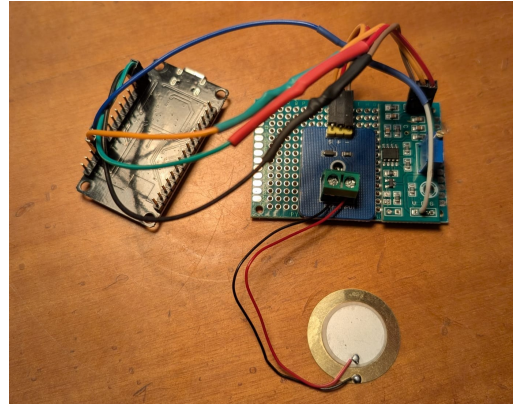
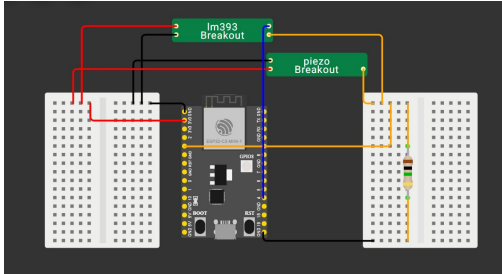
The MCU is **off by default**.
The comparator never sleeps.





Wiring & Photo

3 Energy & Duty Cycle





Wake-up: LM393 $\rightarrow$ TLC393

4 Improvements

Initial choice

- LM393 comparator
- 2.5 mA deep sleep

Result

- TLC393 comparator
- 11 μ A deep sleep
- **227 $\times$ less current**

$$2.5 \text{ mA} \xrightarrow{\text{swap}} 11 \mu\text{A}$$



Protocol: BLE → ESP-NOW

4 Improvements

Initial choice

- BLE for local comms
- ~ 500 ms pairing
- ~ 7.5 ms TX latency

Result

- ESP-NOW
- **No pairing overhead**
- **< 2 ms TX**

Shorter radio-on time = **less energy per alarm**



Acquisition: Polling → Continuous DMA

4 Improvements

Initial choice

- Polling ADC reads
- CPU blocked 100%
- Missed samples at speed

Result

- DMA continuous mode
- **20 kHz**, CPU free

CPU free during capture = **FFT runs in parallel**



MCU: ESP32 V1 → ESP32-C3 RISC-V

4 Improvements

Initial choice

- ESP32 DOIT DevKit V1 dual-core
- Higher idle current: ~ 4 mA (deep sleep)
- Oversized for the task

Result

- ESP32-C3
- $50 \mu\text{A}$ deep sleep
- 20 mA active

Right-sized MCU = months of battery life



Duty Cycle – Energy Calculation (Post-improvements)

4 Improvements

Post-improvement parameters

- Deep sleep: MCU $50 \mu\text{A}$ + TLC393 $11 \mu\text{A} \rightarrow I_{\text{sleep}} = 61 \mu\text{A}$
- ESP32-C3 active baseline: $I_{\text{active}} = 20 \text{ mA}$
- Peak TX: 200 mA (1.5 ms per alarm), 288 alarms/day
- Duty cycle: $D = 10\%$

Average current

$$I_{\text{avg}} = D \cdot I_{\text{active}} + (1 - D) \cdot I_{\text{sleep}} = 0.10 \times 20 \text{ mA} + 0.90 \times 0.061 \text{ mA} \approx 2.06 \text{ mA}$$

Estimated battery life

$$t_{\text{life}} = \frac{2600 \text{ mAh}}{2.06 \text{ mA}} \approx 1262 \text{ h} \approx 52.6 \text{ days}$$

Uses measured/spec'd post-improvement values; TX peaks are negligible on the daily average.



KnockKnock *Thank you for listening!*

Any questions?